SquarePoint

Campus: IIT Delhi · Role: Desk Quant Analyst (DQA)

1. IIT Delhi — Desk Quant Analyst (DQA)

Q1. Top Three Hashtags

You are given n tweets: `tweet_messages[i]` was posted at time `timestamps[i]`. You are also given `current_time` and `window_size`.

Consider only the tweets posted within the time window of length `window_size` that ends at `current_time`. From these tweets, extract all **hashtags**: a hashtag is a `#` followed by one or more non-space characters.

- Hashtags are **case-insensitive**: `#Ronaldo` and `#ronaldo` are the same hashtag.
- Only hashtags that occur **at least twice** are considered.

RANKING

1. Higher frequency first.
2. Tie → the hashtag whose first occurrence is in an earlier tweet (smaller index) comes first.
3. Still tied → lexicographic order of the hashtags.

Return the top three hashtags (fewer if fewer than three qualify).

```
vector<string> topThreeHashtags(vector<string> tweet_messages, vector<int> timestamps,
                                int current_time, int window_size)
```

> Practice (closest): LeetCode 692 — Top K Frequent Words

Q2. Red or Blue

There are n cities on a route after the origin city 0, and two bus services between consecutive cities: **red** and **blue**. $red[i]$ and $blue[i]$ are the times to travel from city $i-1$ to city i by the red and the blue bus respectively.

- You may switch between the red and blue lanes at any city.
- Starting on the blue lane, or switching from red to blue, costs an extra `blueCost` each time. Travelling on red, or switching from blue to red, has no extra cost.

You start at the origin city 0. Return an array ans of size $n+1$, where $ans[i]$ is the minimum time to reach city i, and $ans[0] = 0$.

> Practice (closest): LeetCode 2361 — Minimum Costs Using the Train Line